Ernesto Luis Malta Rodrigues Filho

Software Engineer | C++, C, Java, Python | Azure, OpenShift, Kubernetes | High Performance, Embedded Systems
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Summary

Software Engineer specialized in C/C++ systems engineering and cloud microservices, with 4 years of enterprise energy-sector experience building scalable micro-services and 9+ years of expertise in embedded systems and high-performance applications. Accustomed to working with multinational teams across the USA, Thailand, India, and others. Delivered a configurable data service that eliminated microservice overhead for internal teams and external clients. Developed and maintains C++ open source libraries and projects for desktop and embedded platforms. Driven by complex problem-solving and clean software design, seeking challenging opportunities across backend services, systems, desktop, or embedded platforms.

Experience

Software Engineer

ExxonMobil | Paraná, Brazil *01/2023 – 09/2026*

Developed a centralized, configurable data-access service on a multinational team on Azure Red Hat OpenShift (ARO) that enables clients to deploy customized API endpoints through low-code YAML configurations in GitHub, providing self-service access to Snowflake data without requiring them to build and maintain their own microservices.

• Reduced costs by around **$30k+** per client by centralizing resources on this one-size-fits-all solution, which freed multiple namespace instances costs on Azure, and enabled a more standardized, secure, and performant solution.

• Engineered the supporting CI/CD automation using GitHub Actions, Python, and Bash to validate configurations and streamline deployment of customer-specific service instances.

• Designed the one-size-fits-all API automation approach after researching multiple languages and paradigms, and mentored the distributed USA, Thailand, India and Brazil team through delivery.

• Led improvements to production observability by designing custom telemetry, Datadog monitors, and Snowflake dashboards, providing real-time visibility into pipeline health and operational failures.

• Spearheaded cross-platform build standardization across Windows, MacOS and Linux, eliminating the need for additional hardware and avoiding **$120k+ in costs**.

Skills: C++, Java, Spring Boot, Cloud, Azure, TDD, Redis, REST, SQL, Snowflake, JWT, OAuth2, Kubernetes, k6, CI/CD, Git, Python, GitHub Actions, Datadog, Windows, Linux

Projects & Open Source

Lunaris Library Collection

github.com/LunarisLib *10/2020 – Present*

Authored a cross-platform, open source, utility library featuring lock-free thread primitives, asynchronous I/O, socket abstraction, encryption, and a memory-optimized JSON parser based on cJSON for embedded devices.

• Engineered a suite of 14+ modern C++ modules to accelerate application development, centralizing core features to eliminate code duplication across multiple projects.

• Implemented robust build and testing pipelines using CMake and CTest, which reduced compilation times, standardized workflows, and simplified library linking for dependent applications.

• Created automated version release pipelines to enable the continuous deployment of libraries, simplifying dependency linking for downstream applications.

Skills: C++, CMake, Git, GitHub Actions, Shell, Bash, CI/CD, Windows, Linux, Automated Testing, Clean Architecture

Lohk's Studios Works Engine (multimedia library)

github.com/LohksStudiosWorks/LSWEngine *10/2026 – Present*

Engineered a modern C++20 cross-platform multimedia framework built on top of Allegro 5, abstracting low-level C APIs into safe, high-level abstractions using RAII, thread-safe primitives, and lazy-initialized singletons for deterministic lifecycle management.

• Designed automated cross-platform build and test pipelines using modern CMake and CTest, enabling monolithic builds, rapid incremental compilation, and continuous integration via GitHub Actions.

• Expanded core platform capabilities by implementing native socket networking, parallel processing algorithms, custom collision detection, extended UI dialog tooling, and expanded media format support.

Skills: C++, C, CMake, Git, GitHub Actions, Shell, Bash, CI/CD, Windows, Linux, Automated Testing, Clean Architecture

Modular Monitor Project

Capstone Project – github.com/ModularMonitor *01/2024 – 10/2024*

Developed a modular real-time monitoring engine for ESP32 in C++ with hot-swappable I²C sensor architectures, multithreading, and custom device drivers.

• Developed drivers for hot-swappable Serial through I²C communication, custom TCP socket design for wireless monitoring with REST API, and robust SD card streaming of CSV data.

• Enhanced open source TFT display library for responsive touchscreen user interface, 4G GSM connection, Circular Queue, and multiple Adafruit sensor libraries, to fit constrained hardware.

• Enabled multi-sensoring multi-module design on Espressif ESP32 platform by leveraging multiple hardware-specific tweaks.

Skills: C++, ESP-IDF, Arduino, CMake, Shell, Bash, CI/CD, Git, REST

Personal website

github.com/Lohkat – lohkat.github.io *06/2023 – Present*

Developed and deployed a responsive web portfolio made with modern HTML5, CSS3, and JavaScript, utilizing GitHub Pages as backend with automated deployment and modular design.

• Built interactive frontend features – dynamic galleries, web-player, and search tools – showcasing cross-domain technical adaptability.

• Architected modular design by utilizing multiple repositories, spreading weight and deployments for different parts of the website, accomplishing performance on updates and releases.

Skills: HTML, CSS, JavaScript, Git, GitHub Actions, CI/CD

Education

M.S. in Electrical Engineering & Industrial Informatics (CPGEI) *01/2026 – Present*
Universidade Tecnológica Federal do Paraná – Paraná, Brazil

B.S. in Electrical and Electronics Engineering *01/2016 – 10/2024*
Universidade Tecnológica Federal do Paraná – Paraná, Brazil

Languages

English: Full professional proficiency.
Portuguese: Native

Skills

Languages: C++, C, Java, Shell / Bash, Python, JavaScript / TypeScript, HTML, CSS
Systems & Infrastructure: Linux, Windows, FreeRTOS, Azure, Red Hat OpenShift, Docker, Kubernetes, GitHub Actions
Tools, Frameworks & Practices: CMake, Git, Spring Boot, ESP-IDF, Arduino, REST, Snowflake, Redis, Datadog, k6, CI/CD, TDD (Test Driven Development), Event Driven, Agile / Scrum, Clean Code , automated testing